



DEPARTMENT OF COMPUTER ENGINEERING & APPLICATIONS
Institute of Engineering & Technology

Lab Manual

**Subject Name & Code: OPERATING SYSTEM LAB
(BCAC 0818)**

Course: BCA

Faculty's Name: **Dr. Anuj Mangal**
Dr. Anil Kumar Chanchal
Ms. Paromita Goswami
Mr. Anuj Kumar
Mr. Santosh Swarnkar
Mr. Jayati Krishna Goswami
Ms. Cheshtha Bharadwaj

Year: II

Semester: III

Session: 2024-25

OPERATING SYSTEM LAB (BCAC 0818)

Prerequisite: Familiarity with programming constructs

Credits: 02

Semester III

L-T-P : 0-0-2

Prac. No.	Program	Page No.
1	Case Study On Open Standard and Software: Introduction to Linux operating system and Preparing for Installation – Installation Checklist, Hardware Requirements, Partitioning, Installation problems, Working with the System, Shells and Utilities, Linux commands, File Handling using vi editor, Getting familiar with shell scripts.	
2	Implement the following basic commands(with options) used in UNIX/LINUX OS: ls, pwd, tty, cat, who, who am I, rm, mkdir, rmdir, touch, cd.	
3	Implement the following basic commands(with options) used in UNIX/LINUX OS: cal, cat(append), cat(concatenate), mv, cp, man, date.	
4	Implement the following basic commands(with options) used in UNIX/LINUX OS: chmod, grep, tput (clear, highlight), bc.	
5	Write a shell script to check if the number entered at the command line is prime or not.	
6	Write a shell script to modify the “cal” command to display calendars of the specified months.	
7	Write a shell script to modify the “cal” command to display calendars of the specified range of months.	
8	Write a shell script to accept a login name. If not, a valid login name display message – “Entered login name is invalid”.	
9	Write a shell script to display the date in the mm/dd/yy format.	
10	Write a shell script to display on the screen the sorted output of the “who” command along with the total number of users.	
11	Write a shell script to display the multiplication table of any number.	
12	Write a shell script to compare two files and, if found equal, ask the user to delete the duplicate file.	
13	Write a shell script to check whether the file has all the permissions or not.	
14	Simulate FCFS CPU scheduling algorithm in C++.	
15	Simulate SJF CPU scheduling algorithm in C++.	
16	Simulate Priority CPU scheduling algorithm in C++	
17	Simulate Round Robin CPU scheduling algorithm in C++.	
18	Simulate FIFO page replacement algorithm in C++.	
19	Simulate LRU page replacement algorithm in C++.	